

Algorithms for Programming Contests - Week 12

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Geometry

Classical geometry

- Points as (x, y) coordinates.
- Lines represented by two points, or e.g. with slope intercept form.

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Not many problems, but can be solved! :)

Projective geometry

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Projective Plane

Definition

The *projective plane* is defined as $\mathbb{RP}^2 := (\mathbb{R}^3 \setminus \{0\}) / \sim$, where $\sim$ is an equivalence relation on $\mathbb{R}^3 \setminus \{0\}$ defined by

$$x_1 \sim x_2 :\Leftrightarrow \exists \lambda \in \mathbb{R} : x_1 = \lambda x_2.$$

Its equivalence classes are given by

$$[x] := [x]_{\sim} = \{ \lambda x \mid \lambda \in \mathbb{R} \setminus \{0\} \}.$$

(De-)Homogenization

Note: $\mathbb{R}^2$ can be *embedded*¹ into $\mathbb{RP}^2$ via

$$\mathcal{H} : \mathbb{R}^2 \rightarrow \mathbb{RP}^2, \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \left[\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \right].$$

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This embedding is also called *Homogenization*. The map

$$\mathcal{D} : \{ (x, y, z) \in \mathbb{R}^3 \mid z \neq 0 \} \rightarrow \mathbb{R}^2, \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} x/z \\ y/z \end{pmatrix}$$

is called *Dehomogenization*, as it is also well-defined as map $\mathcal{H}(\mathbb{RP}^2) \rightarrow \mathbb{R}^2$ and inverse to $\mathcal{H}$.

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- Consider the point $(x, y) \in \mathbb{R}^2$, and scale it by a factor of $\lambda > 0$ to obtain $(\lambda x, \lambda y)$.

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- Consider the point $(x, y) \in \mathbb{R}^2$, and scale it by a factor of $\lambda > 0$ to obtain $(\lambda x, \lambda y)$.
- Note that $\mathcal{H}(\lambda x, \lambda y) = [(\lambda x, \lambda y, 1)] = [(x, y, \frac{1}{\lambda})]$, and $(x, y, \frac{1}{\lambda}) \xrightarrow{\lambda \rightarrow \infty} (x, y, 0)$.

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- $[(x, y, 0)]$ is called a *point at infinity*.

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- A *line* in $\mathbb{RP}^2$ is some $[(a, b, c)] \in \mathbb{RP}^2$.
- A point $[(x, y, z)]$ is *incident* to the line $[(a, b, c)]$ iff $\langle (x, y, z), (a, b, c) \rangle = 0$.

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- Note that $[(x, y, z)]$ is incident to $[(0, 0, 1)]$ iff $z = 0$, i.e. the points incident to the line at infinity are exactly the points at infinity.

Cross product

Reminder: given $v_1 = (x_1, y_1, z_1)$, $v_2 = (x_2, y_2, z_2) \in \mathbb{R}^3$, the *cross product* is defined as

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \times \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} := \begin{pmatrix} x_2 y_3 - x_3 y_2 \\ x_3 y_1 - x_1 y_3 \\ x_1 y_2 - x_2 y_1 \end{pmatrix}$$

and is perpendicular to both v_1 and v_2 , i.e.

$$\langle v_1, v_1 \times v_2 \rangle = 0 = \langle v_2, v_1 \times v_2 \rangle.$$

Moreover, if v_1 and v_2 are not linearly dependent, $v_1 \times v_2$ is unique (up to scaling) with this property.

Operations in $\mathbb{RP}^2$

Intersection of two lines



Line through two points



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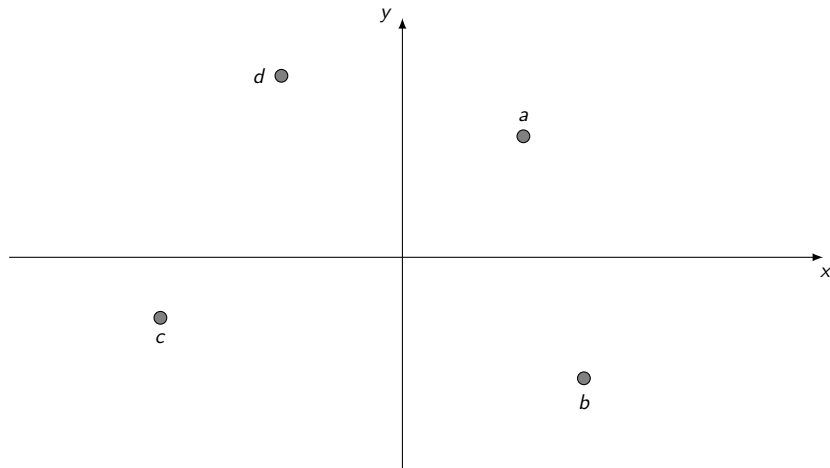
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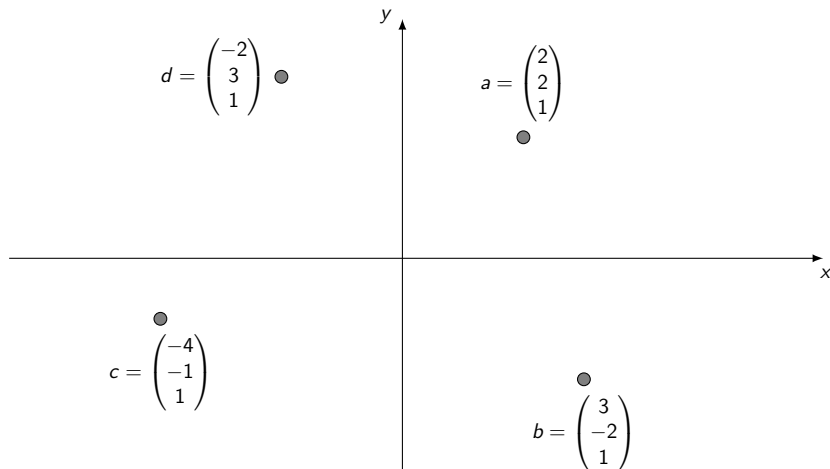
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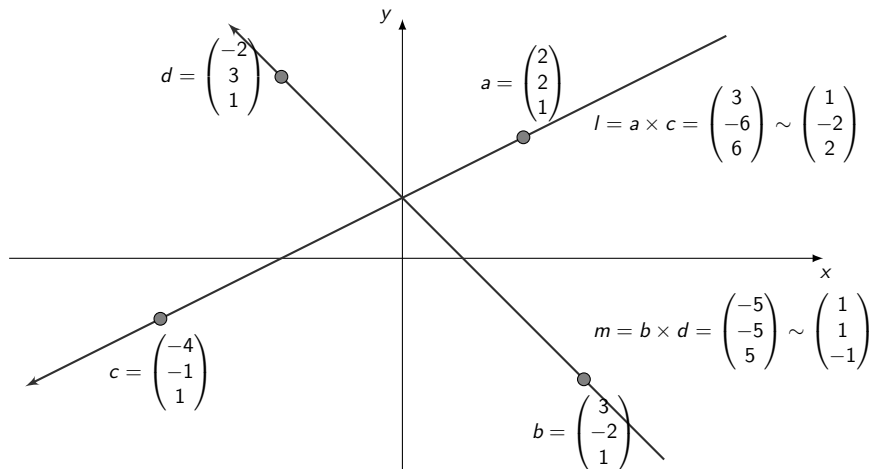
Example: intersection of diagonals



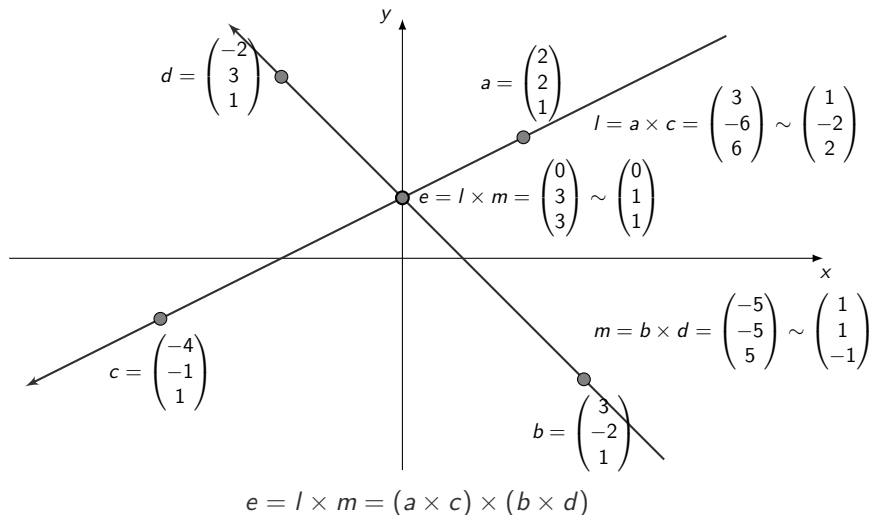
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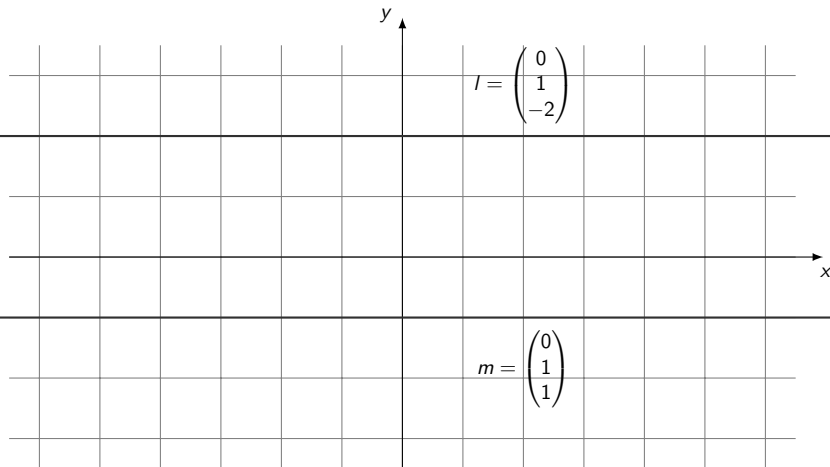
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- In $\mathbb{RP}^2$, every two distinct lines intersect.
 - If two lines are "parallel", their intersection is a point at infinity.
 - The intersection of any of the parallel lines with the line at infinity is exactly this point at infinity.

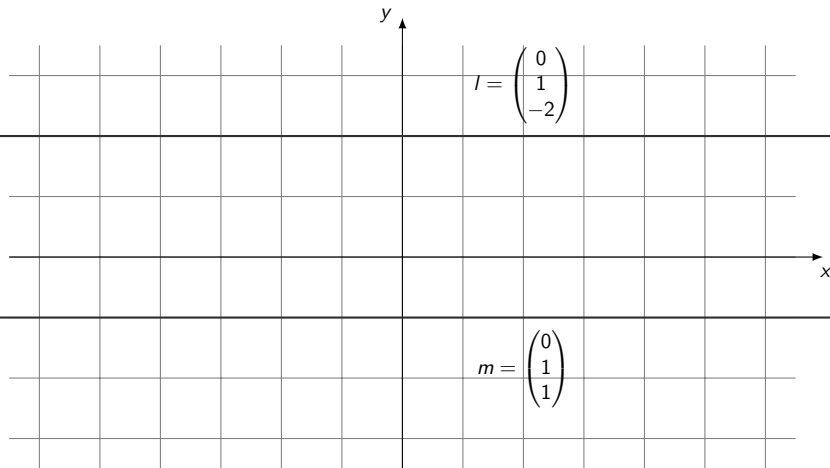
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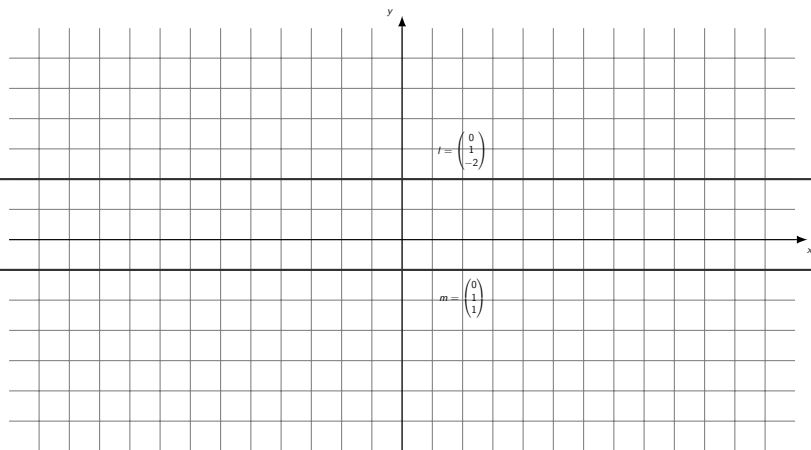
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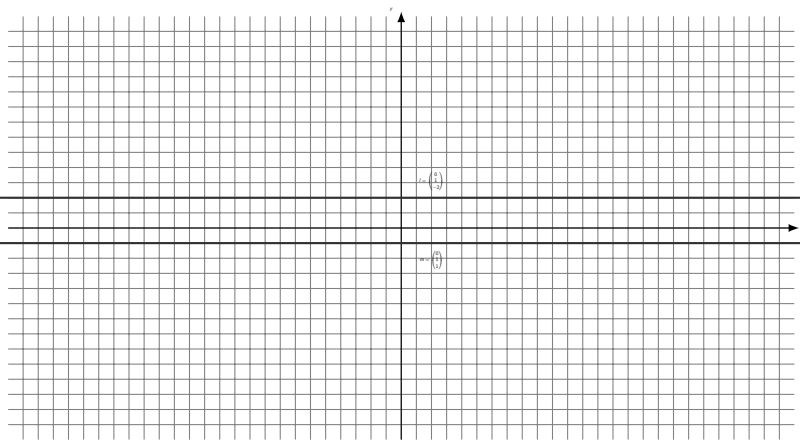
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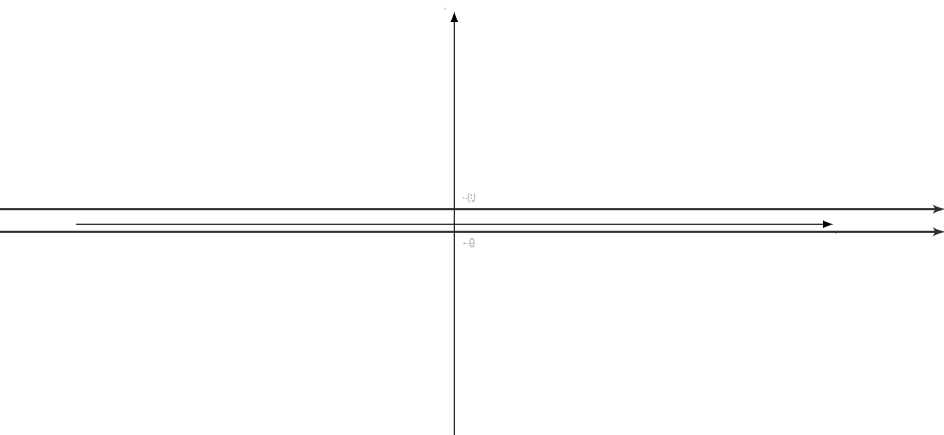
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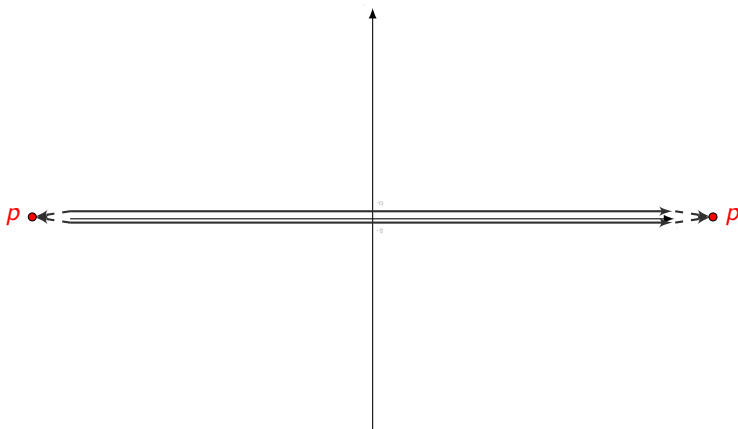
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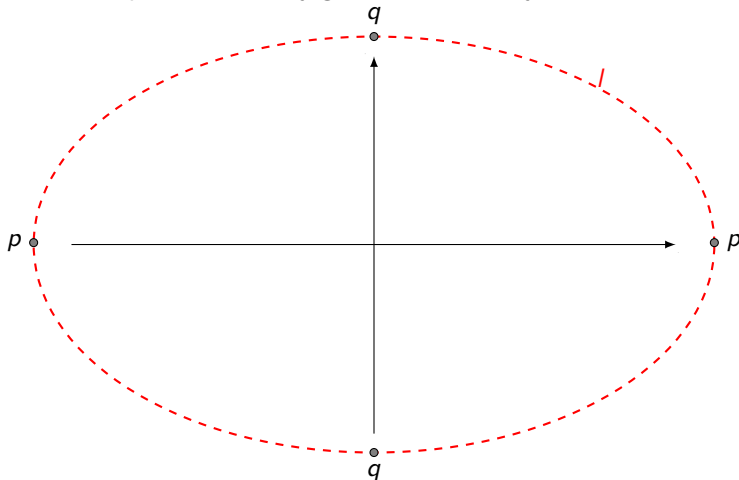
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Points at infinity revisited

Connection of points at infinity gives line at infinity:



$$l = p \times q = (1, 0, 0)^T \times (0, 1, 0)^T = (0, 0, 1)^T$$

Constructions with points at infinity

Let $[l_1]$, $[l_2]$ be two lines, both not the line at infinity. $[l_1]$ and $[l_2]$ are *parallel* iff their intersection is a point at infinity.

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Parallel

- Goal: construct line $[m]$ parallel to $[l]$ through $[p]$.
- Point at infinity in direction of $[l]$: $[q] = [l \times (0, 0, 1)]$. Then $[m] = [p \times q]$.

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Perpendicular

- Goal: construct line $[m]$ perpendicular to $[l]$ through $[p]$.
- Point at infinity in direction of $[l]$: $[q] = [l \times (0, 0, 1)] =: [(x, y, 0)]$.
- Orthogonal direction to q : $q^\perp = (y, -x, 0)$. Then $[m] = [p \times q^\perp]$.

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Projection

- Goal: Construct projection $[r]$ of $[p]$ on $[l]$.
- Compute perpendicular $[m]$, then $[r] = [m \times l]$.

Transformations in Euclidean and projective space

Rotations

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Translations

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$

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In projective space

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

Affine transformations

Transformations preserving parallelity

- Shifting
- Scaling
- Rotating
- Mirroring
- Shearing
- Combinations of all these operations

$$M = \begin{pmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{pmatrix}, \det(M) \neq 0$$

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \mapsto M \cdot \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

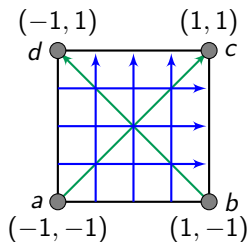
Projective transformations

Arbitrary projective transformation

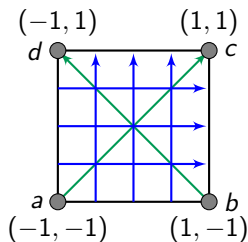
$$M = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}, \det(M) \neq 0 \qquad \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto M \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

- M and λM for $\lambda \neq 0$ represent the same transformation.
- All affine transformations and perspective distortions.
- Collinearity and point-on-line property is preserved.
- Every transformation preserving collinearity is projective.
- Uniquely determined by four non-collinear points and their images.

Example: projective transformation

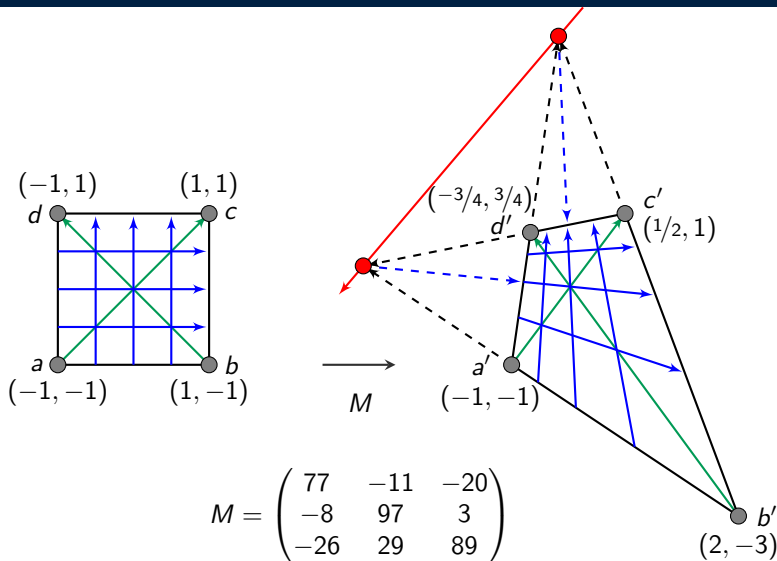


Example: projective transformation



$$M = \begin{pmatrix} 77 & -11 & -20 \\ -8 & 97 & 3 \\ -26 & 29 & 89 \end{pmatrix}$$

Example: projective transformation



Computing projective transformations

Computing M

Given four points a, b, c, d and their respective images a', b', c', d' by a projective transformation, what is a matrix M for this transformation?

Assumption: no three points of a, b, c, d or a', b', c', d' are collinear.

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Solve linear equation system to find M and $\lambda_a, \lambda_b, \lambda_c, \lambda_d$ such that:

$$M \cdot a = \lambda_a a' \quad M \cdot b = \lambda_b b' \quad M \cdot c = \lambda_c c' \quad M \cdot d = \lambda_d d'$$

M unique up to scalar multiple: one λ can be fixed.

Solve linear equation system with $9 + 3 = 12$ variables and 12 equations.

Computing projective transformations

Fix $\lambda_d = 1$, and consider the following special case:

$$a = (1, 0, 0)^T \quad b = (0, 1, 0)^T \quad c = (0, 0, 1)^T \quad d = (1, 1, 1)^T$$

$$M \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \lambda_a a' \quad M \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \lambda_b b' \quad M \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \lambda_c c' \quad M \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = d'$$

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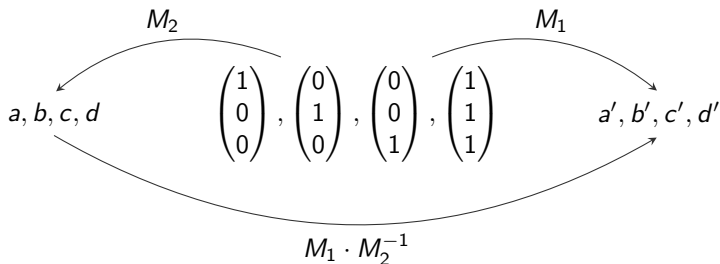
$$M \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \lambda_a a' \quad M \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \lambda_b b' \quad M \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \lambda_c c' \quad M \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = d'$$

$$M = \begin{pmatrix} | & | & | \\ \lambda_a a' & \lambda_b b' & \lambda_c c' \\ | & | & | \end{pmatrix} \quad \begin{pmatrix} | & | & | \\ a' & b' & c' \\ | & | & | \end{pmatrix} \cdot \begin{pmatrix} \lambda_a \\ \lambda_b \\ \lambda_c \end{pmatrix} = d'$$

Only need to invert single 3×3 matrix to obtain $\lambda_a, \lambda_b, \lambda_c$ and then M .

Computing projective transformations

General case:



More projective geometry

There is a lot more

- Circles, conic sections.
- Measurements of lengths and angles.
- Cross-ratios and projective invariants.
- Higher-dimensional projective geometry.
- $\Rightarrow$ lectures “Geometriekalküle” and “Projective Geometry” by Prof. Richter-Gebert.